

WADENA, MN, USA

WMO: 726561

Lat: 46.447N

Lon: 95.212W

Elev: 1370

StdP: 13.98

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 94997

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.2	-16.9	-26.5	1.3	-20.6	-21.8	1.7	-15.6	25.2	11.6	22.9	11.5	4.9	260	0.620

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	86.0	71.6	82.2	69.7	80.9	68.4	74.8	82.7	72.5	80.0	70.2	77.6	11.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.3	126.0	80.0	70.2	116.8	77.7	67.6	106.8	75.0	39.2	83.0	37.1	79.5	35.1	76.1	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.4	19.5	17.6	DB	-28.4	91.1	5.1	3.1	-32.0	93.3	-35.1	95.1	-37.9	96.8	-41.7	99.0	(4)
(5)			WB	-28.6	77.6	5.1	2.5	-32.3	79.3	-35.3	80.8	-38.1	82.2	-41.8	84.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.9	8.5	10.7	25.9	40.4	52.8	62.7	67.7	64.8	57.5	43.0	28.3	14.2		
	DBStd	23.24	14.17	12.77	13.38	10.43	8.86	6.36	5.69	5.98	8.77	9.21	10.87	12.48		
	HDD50	5571	1286	1101	750	313	70	2	0	1	28	258	653	1109		
	HDD65	9454	1751	1521	1212	738	391	114	37	76	255	683	1102	1574		
	CDD50	1874	0	0	4	25	157	383	550	460	252	42	1	0		
	CDD65	281	0	0	0	0	13	45	122	71	29	1	0	0		
	CDH74	2804	0	0	0	16	177	439	1147	697	305	23	0	0		
	CDH80	661	0	0	0	4	42	84	291	167	69	4	0	0		
	WSAvg	7.6	7.7	7.6	8.1	8.9	8.8	7.2	6.1	5.9	7.2	7.8	7.9	7.5		
	PrecAvg	27.8	0.8	0.7	1.4	2.4	3.3	4.4	3.9	3.0	2.9	3.0	1.3	0.7		
(15) Precipitation	PrecMax	33.9	2.8	2.0	3.6	4.9	7.1	7.3	8.3	7.4	6.7	6.9	4.0	1.7		
	PrecMin	21.8	0.1	0.2	0.5	1.0	0.7	1.7	1.3	0.7	1.0	0.3	0.0	0.3		
	PrecStd	3.8	0.6	0.5	0.7	1.1	1.5	1.6	1.7	1.9	1.3	1.9	1.0	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.2	45.0	64.1	73.3	84.4	87.8	90.9	89.8	87.7	76.9	61.3	43.3		
		MCWB	35.2	41.0	54.2	58.2	66.1	73.1	75.7	74.5	71.4	61.2	50.6	39.2		
	2%	DB	34.4	37.2	56.5	67.6	79.3	81.9	86.0	83.9	81.4	70.0	54.2	36.7		
		MCWB	32.1	34.3	48.5	53.6	63.0	69.0	73.1	71.3	68.5	58.2	46.7	34.0		
	5%	DB	30.3	34.1	48.5	63.2	73.2	79.2	82.3	81.3	77.4	63.2	47.7	33.8		
		MCWB	29.6	32.1	43.7	50.0	59.6	65.9	70.9	69.0	65.5	54.3	42.6	32.2		
	10%	DB	27.9	29.6	43.4	57.3	69.8	75.3	81.0	78.7	72.7	57.0	43.3	30.1		
		MCWB	26.8	28.6	38.3	46.6	58.0	64.3	69.4	67.0	63.9	49.9	38.7	29.5		
	0.4%	WB	35.8	41.1	56.4	59.6	69.1	76.2	78.4	76.2	74.4	64.1	52.9	39.3		
		MCDB	39.7	44.6	61.6	75.0	80.7	83.6	86.2	84.8	82.2	72.6	59.1	42.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.3	35.1	49.5	54.4	65.2	71.9	75.5	73.3	70.6	59.3	47.1	34.4		
		MCDB	34.3	37.6	54.4	64.3	74.2	78.6	83.5	80.8	78.2	67.9	52.1	36.4		
	5%	WB	29.8	31.8	44.3	51.1	62.2	68.8	73.2	71.0	67.9	54.7	42.9	31.8		
		MCDB	30.7	33.7	49.8	60.8	70.7	76.6	80.9	78.6	74.9	61.1	47.2	33.2		
	10%	WB	27.2	27.9	38.9	48.0	59.0	66.2	70.6	68.6	65.3	51.2	39.2	29.7		
		MCDB	27.7	28.7	43.0	58.1	67.3	73.4	77.9	75.7	70.9	56.4	43.1	30.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.3	18.9	18.3	20.4	21.4	20.2	21.4	21.7	21.5	17.8	15.7	15.0		
		MCDBR	17.1	19.1	24.3	29.6	28.9	24.3	24.0	25.0	26.6	27.0	23.7	17.7		
		MCWBR	14.8	15.9	16.9	18.0	15.9	13.1	12.9	13.2	14.3	16.6	16.3	15.0		
		MCDBR	16.4	17.6	23.7	26.8	24.9	20.6	22.1	21.7	22.4	24.1	21.4	16.4		
	5% WB	MCWBR	14.3	14.7	17.0	17.0	15.0	12.9	13.2	12.6	13.4	15.6	15.4	14.3		
(40) Clear-Sky Solar Irradiance	taub		0.263	0.298	0.302	0.356	0.375	0.378	0.381	0.383	0.352	0.309	0.279	0.258		
	taud		2.362	2.257	2.395	2.328	2.338	2.362	2.371	2.375	2.443	2.544	2.516	2.412		
	Ebn at Noon		268	276	294	285	282	281	279	273	274	273	262	258		
	Edh at Noon		23	31	32	37	38	38	37	36	30	23	20	19		
(44) All-Sky Solar Radiation	RadAvg		511	866	1215	1537	1704	1870	1986	1671	1251	774	479	353		
	RadStd		59	92	120	127	125	124	84	145	94	119	49	45		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page